

Overview

My research addresses a single structural problem that cuts across AI security, AI safety, and AI evaluation: **standard evaluation metrics systematically misrepresent what AI models are actually doing**. This is not a claim about measurement noise. It is a claim about structural properties of how current evaluation pipelines are designed with properties that cause them to conflate qualitatively distinct behaviours, present contested judgments as ground truth, and remain blind to causally verified failure modes. I have demonstrated this problem empirically across three domains i.e. multilingual LLM safety evaluation, hate speech annotation, and medical Vision Transformers and I am pursuing its implications for AI agent safety and alignment.

This research connects directly to the COMPASS group's work on probing and evaluating the failure modes of AI models, designing mitigations and white-box control methods, and understanding the human and societal aspects of AI deployment. My work provides concrete empirical grounding for the theoretical concerns about evaluation reliability that motivate much of COMPASS's research programme. I write this research statement based on how I have progressed on AI safety in the 3 research threads I have been involved in.

Research Thread 1: Evaluation Metrics Conflate Distinct Failure Modes

Multilingual LLM Safety Evaluation (ACL submission)

Binary refusal-rate metrics are the standard tool for multilingual LLM safety evaluation. A model that refuses 100% of English harmful prompts but only 6% of Luganda prompts is reported as dramatically less safe in Luganda. My work shows this conclusion is wrong and that the wrongness is structural, not incidental.

I evaluated three instruction-tuned open LLMs (Gemma-2-2B-IT, Llama-3-8B-Instruct, Qwen-2.5-7B-Instruct) on 30 AdvBench harmful behaviours translated into Swahili, Yoruba, and Luganda, then manually triage all 207 non-refusal completions. Of 207 cases flagged as unsafe by the naive metric, only 2 contained coherent harmful content (0.97%). The remaining 205 are distributed across three failure modes the binary metric conflates: *capability collapse* (model fails to parse the prompt), *hallucinated translation* (model invents a benign meaning), and *pseudo-compliance*

(model adopts the harmful framing but produces semantically empty output). Each model exhibits a different dominant mode ($\chi^2 = 89.4$, $df = 4$, $p < 0.0001$).

The 30 to 80 percentage point gap between naive and triaged rates is not measurement noise. It is a structural property of how current evaluation collapses qualitatively distinct behaviours into a single number. This connects directly to COMPASS's work on understanding the failure modes of AI models and to the broader question of what evaluation pipelines are actually measuring.

Research Thread 2: Aggregation Presents Contested Judgments as Ground Truth

Majority Vote and Annotator Disagreement (ICML 2026, Pluralistic Alignment Workshop)

Standard hate speech annotation pipelines collapse annotator disagreement into majority vote labels before training. My work shows this aggregation is not neutral: 42.6% of all annotator disagreement in HateXplain concentrates specifically at the hate/offensive boundary ($\chi^2 = 135.199$, $df = 2$, $p < 0.0001$). This pattern is consistent with annotators applying different thresholds for where hate begins i.e. a value judgment, not a factual dispute and majority vote resolves it by silencing whichever annotators land on the minority side.

Three BERT models of increasing sophistication (hard labels, soft labels, per-annotator heads) all fail to recover boundary accuracy. The agreement gap does not shrink under stronger models instead it widens (from 22.2 to 28.1 percentage points). The only consistent improvement is calibration: hard-label models are significantly overconfident on boundary case errors (0.710 vs. 0.495 for per-annotator models, $p < 0.0001$), meaning standard accuracy metrics will not detect the failure. A model that is wrong and certain on culturally contested inputs is a dangerous deployment outcome that current evaluation will systematically miss.

This work provides direct empirical grounding for the pluralistic alignment concern: RLHF pipelines collapse human preference comparisons into reward signals that may encode majority rater values as universal.

Research Thread 3: Output-Level Evaluation Is Blind to Mechanistic Failure

Spurious Attention Heads in Medical Vision Transformers (IJCAI submission)

I applied mechanistic interpretability to a medical Vision Transformer to identify and causally verify a spurious annotation-reading circuit. A systematic sweep of all 144 attention heads in ViT-Base/16 identified L8H10 as a dedicated corner-contrast detector: it places 34.6% of its attention on annotation text in 62% of 200 clinical chest X-rays, with a corner-to-lung attention ratio $74.8\times$ higher in affected images ($p < 0.0001$). Two independent causal interventions confirm annotation content, not luminance or edge density drives the behaviour.

The central finding is not the spurious head itself. It is the evaluation gap: single-head ablation of L8H10 produces $p = 0.340$ on downstream classification. Multi-head ablation of all three identified spurious heads produces $p = 0.532$. A causally verified annotation-reading circuit is completely invisible to output-level evaluation. This finding generalises across DeiT-Base (88% spurious rate) and ViT-Large/16 (80% spurious rate), and persists through full fine-tuning. The spurious rate marginally increases, as the shortcut is reinforced rather than removed.

The implication for AI safety evaluation is direct: standard evaluation answers whether the model performs correctly on this data. Mechanistic auditing answers which components are responsible, and whether any are doing so for the wrong reasons. These questions are not interchangeable. A model can answer the first correctly while failing the second catastrophically and the failure will remain invisible until the distributional shift exposes it. This structural blindness in output-level evaluation is the same problem I document in the safety and annotation threads: the metric was never measuring what it claimed to measure.

Research Thread 4: Adversarial Robustness and Evaluation Under Adaptive Attack

RL-Based IDS Evasion Arms Race (ANUBIS @ ESORICS 2026, under review)

A Random Forest IDS with 100% ROC-AUC is evaded by random perturbation 64.2% of the time. This is the same evaluation reliability problem applied to adversarial robustness: benchmark accuracy is not adversarial robustness, and the metric was never measuring what it claimed to measure.

I implemented a three-round arms race between a PPO evasion agent and an iteratively retrained Random Forest IDS on CICIDS 2018, with random-agent

baselines at every round. The central finding is the divergence between PPO and random perturbation as detectors harden: a random agent collapses from 64.2% to 22.4% evasion across three rounds, while PPO maintains 65.4%, a 43 percentage-point advantage that constitutes empirical evidence of genuine adaptive learning. SHAP analysis identifies the structural reason: the detector persistently over-weights the same TCP connection-setup parameters the agent exploits, and adversarial retraining alone cannot redistribute that weight.

This work establishes methodological discipline that I carry across all my research: explicit baselines at every evaluation point, statistical validation with bootstrap confidence intervals, honest acknowledgment of what the results do and do not prove.

Proposed Research

Sahar Abdelnabi's work on the Hawthorne effect in reasoning models showing that LLMs behave differently during evaluation than during deployment, making standard benchmarks structurally misleading is the same argument I make across my three empirical domains. Her 2023 work on indirect prompt injection identified a failure class that was invisible to standard safety evaluation until attacked from a direction the evaluation was not designed to detect. The COMPASS group's current work on evaluation awareness, multi-agent collusion auditing, and contextual integrity addresses exactly the question I am pursuing: what are AI evaluation pipelines actually measuring, and what dangerous properties remain invisible to them?

The research direction I propose for a PhD within COMPASS extends this agenda to LLM agent evaluation. Specifically: **when LLM agents are evaluated for safety, alignment, and robustness, which failure modes are systematically invisible to the evaluation methodology, and why?** My prior work provides three empirical instances of this question in simpler systems. Extending it to LLM agents introduces additional dimensions: multi-turn interaction, tool use, memory across sessions, and adversarial manipulation of the evaluation environment.

Concrete research questions I would pursue include: how do indirect prompt injection attacks interact with agent evaluation benchmarks (can an adversarial environment cause an agent to perform well on evaluation while exhibiting unsafe behaviour in deployment)? What is the equivalent of pseudo-compliance, structurally correct but semantically empty output in multi-agent coordination tasks? And how does the

aggregation problem I document in annotation translate to RLHF reward models trained on human preference data from demographically non-representative rater pools?

My work in African language NLP and medical AI in low-resource settings also connects to COMPASS's emphasis on the human and societal aspects of AI deployment. The populations most harmed by unreliable AI evaluation metrics are often those in low-resource contexts i.e. African-language speakers whose models are incorrectly declared unsafe, communities whose content moderation outcomes are decided by models trained on majority annotator judgments that do not represent their norms, and clinicians in under-resourced health systems whose diagnostic tools silently transfer shortcuts from high-resource training environments. Building AI evaluation methodology that is reliable for these populations is not a niche concern. It is a prerequisite for AI systems that are genuinely safe and aligned.

Summary

My research demonstrates, across four empirical domains, that standard AI evaluation metrics systematically misrepresent what models are actually doing. The failure is not random: it is structural, it is consistent across architectures and settings, and it preferentially affects the populations who are already underserved by AI systems. The intervention must be upstream in evaluation design and not downstream in model training.

COMPASS is the research group best positioned to extend this agenda to the domain where it matters most: LLM agents deployed in adversarial environments, evaluated by pipelines that may be as structurally blind as the ones I have already documented. I am eager to contribute to that programme.

Selected Papers

- **Refusal Rate Is Not Safety: Three Failure Modes of Multilingual LLM Safety Evaluation.** Anonymous ACL submission, 2026.
- **Majority Vote Silences Minority Values: Annotator Disagreement at the Hate/Offensive Boundary in HateXplain.** Pluralistic Alignment Workshop @ ICML 2026, Seoul, South Korea.
- **Spurious Attention Heads in Vision Transformers: Mechanistic Identification of Annotation-Driven Bias in Medical Chest X-Ray Analysis.** Anonymous IJCAI submission, 2026.
- **Adversarial Evasion of AI-Based Intrusion Detectors Using Reinforcement Learning: An Arms Race Analysis.** Under review, ANUBIS @ ESORICS 2026.